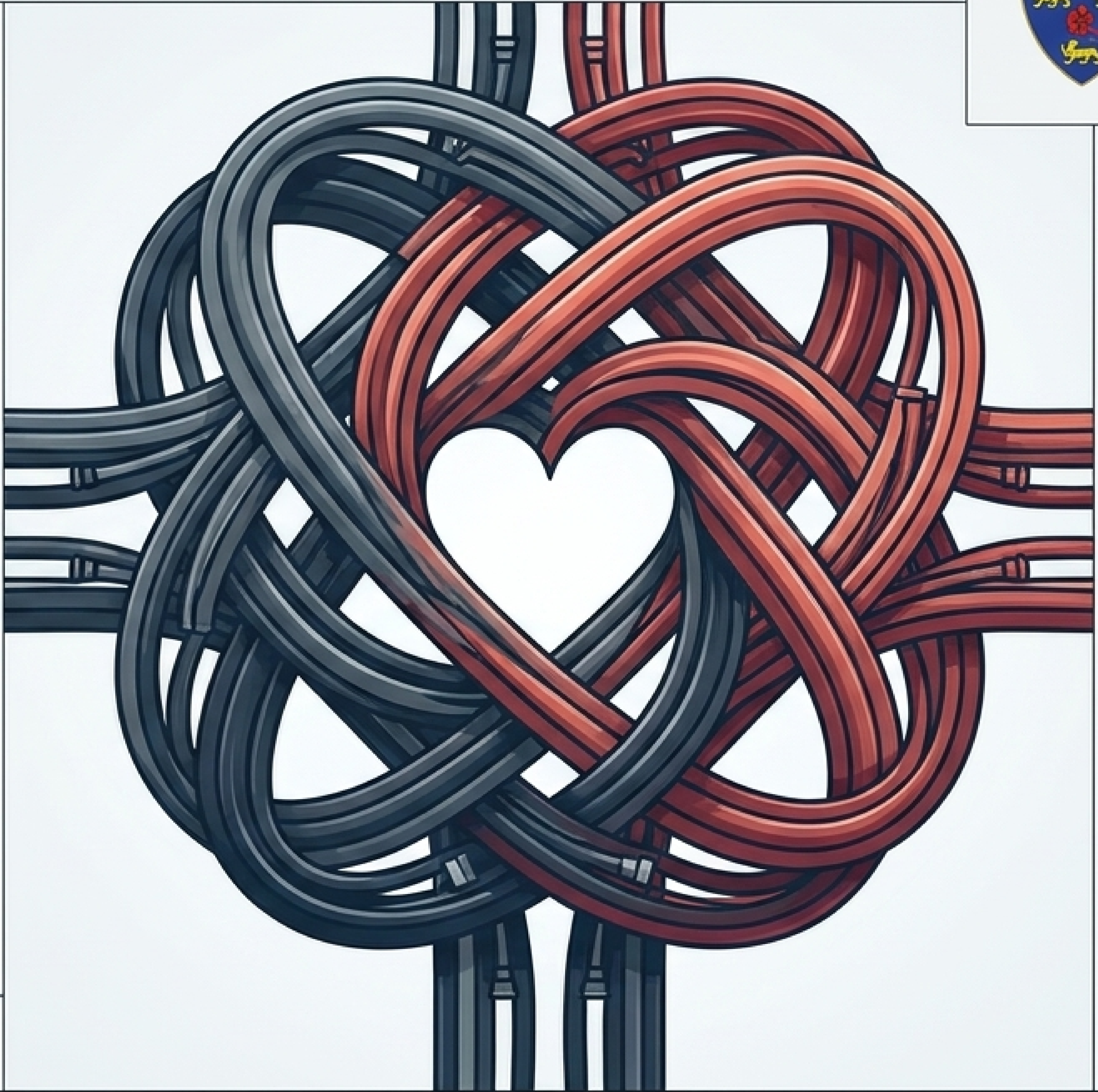
Tying the Data Knot

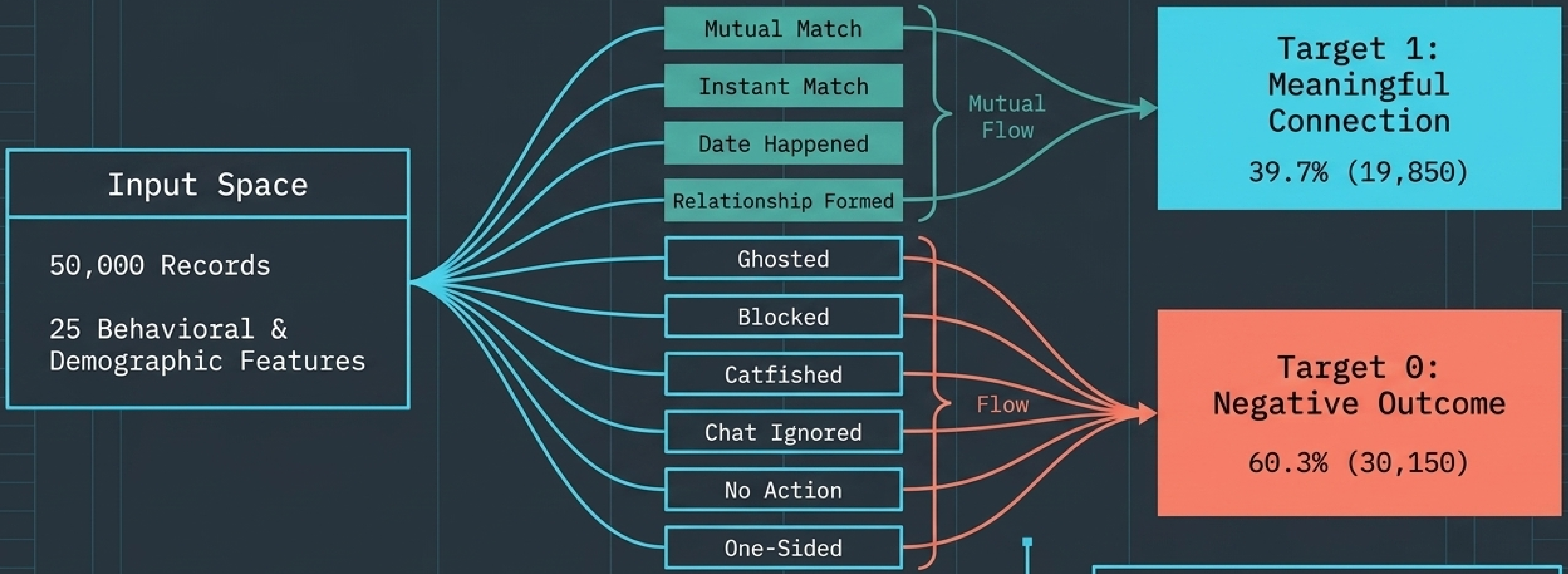
A State-of-the-Art Machine Learning Pipeline for Digital Romance

V8 Patched Edition



OCC6 Group 3 | WIA1006 Machine Learning | University of Malaya

The Dataset & Target Variable



Analytical Takeaway: Simplifies the predictive mathematical objective and yields a stable baseline for SCM causal analysis.

Baseline Benchmarking & AutoML Validation

Establishing performance limits across 14 distinct architectures.

AutoML Objective Validation

PyCaret & FLAML frameworks deployed for unbiased baseline benchmarking.

Custom Architectures Evaluated

Linear/Distance: Logistic Regression, Cosine KNN CF

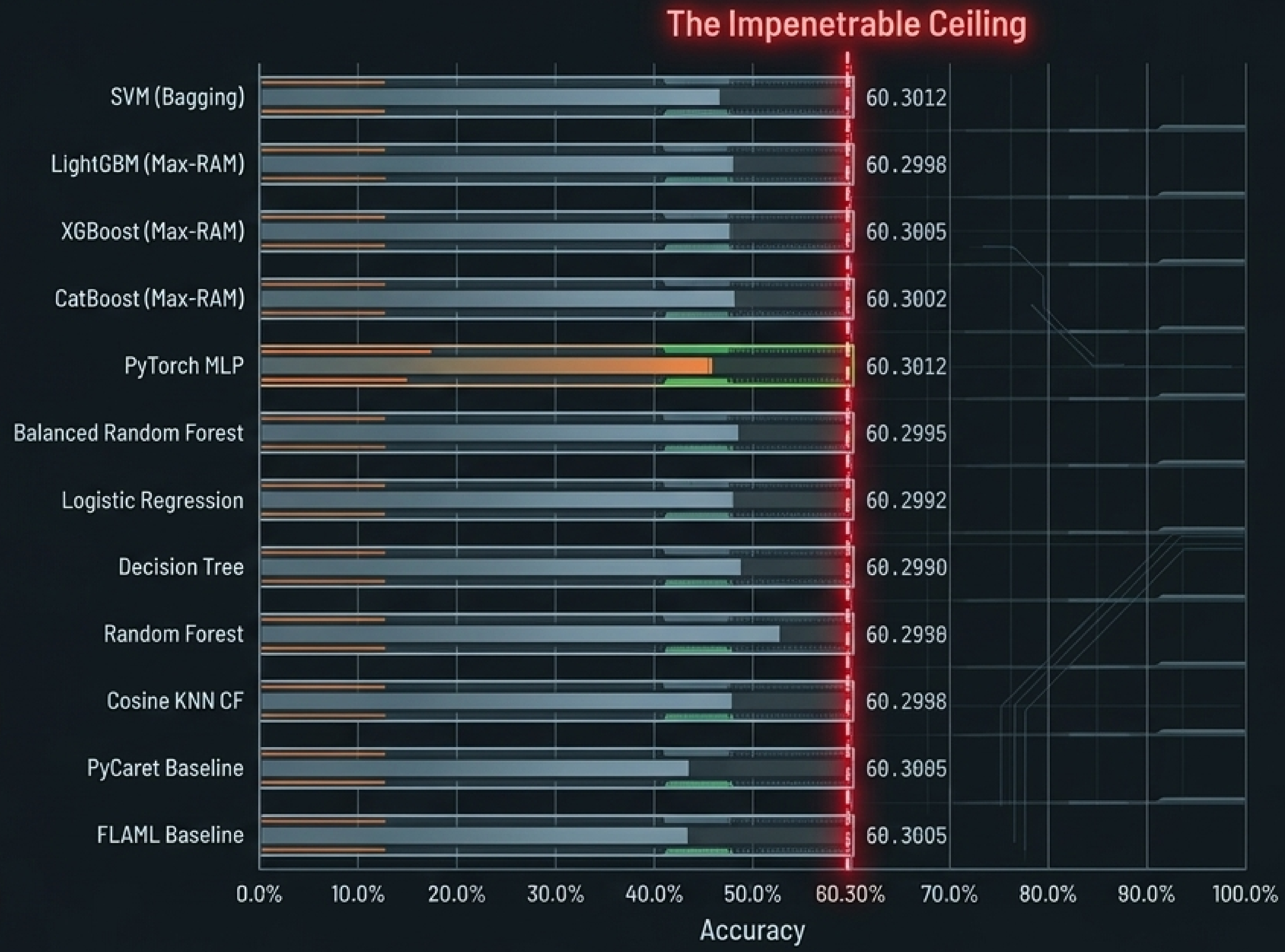
Tree-Based: Decision Tree, Random Forest, Balanced RF

Max-RAM Boosting: XGBoost, LightGBM, CatBoost (Scaled to 1,000 estimators, depth 12)

Compute-Heavy: 16-Thread Bagging SVM, PyTorch MLP

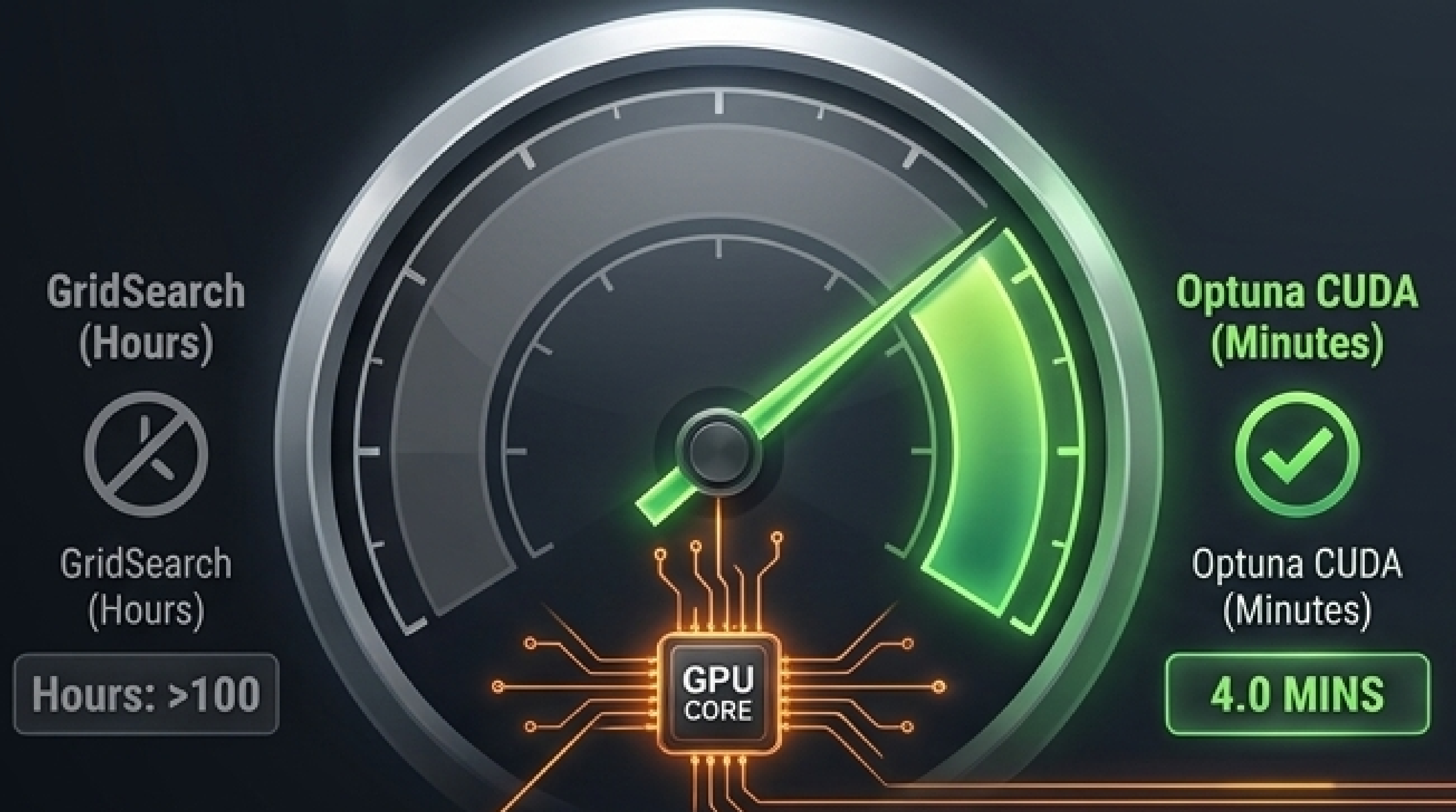
⚠ The Verdict

Despite state-of-the-art engineering, accuracy across all 14 models aggressively converges precisely at the ~60.30% majority baseline.



Massive GPU-Accelerated Optuna Tuning

Offloading hyperparameter trials to CUDA cores for extreme efficiency.

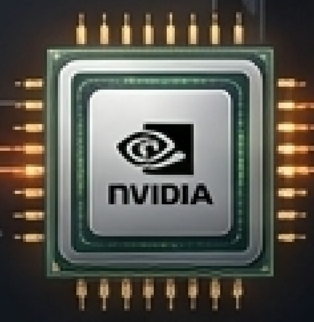



```python
# Massive GPU Offloading with Optuna
import optuna
import torch

study = optuna.create_study(direction="maximize")

# Optimize with GPU acceleration for extreme efficiency
# n_jobs=1 ensures each trial fully utilizes the dedicated GPU
study.optimize(objective, n_trials=1000, n_jobs=1)

# Within objective function:
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
# Ensure model and data are moved to the device for accelerated training
model.to(device)
data, target = data.to(device), target.to(device)
```

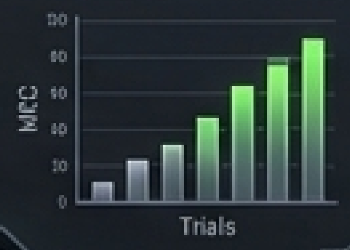


CUDA Cores Activated

The Optimization Engine

Replaced legacy GridSearch with a massive 1,000-trial Optuna framework.

Target Objective: Maximizing Matthews Correlation Coefficient (MCC) to handle residual imbalance.



The Hardware Acceleration

Estimator fitting offloaded directly to dedicated GPU CUDA and OpenCL cores.

Velocity: Trial completion in ~0.1 to 0.2 seconds.

Scale: 1,000 high-complexity tuning iterations executed seamlessly in under 4 minutes.

The Verdict

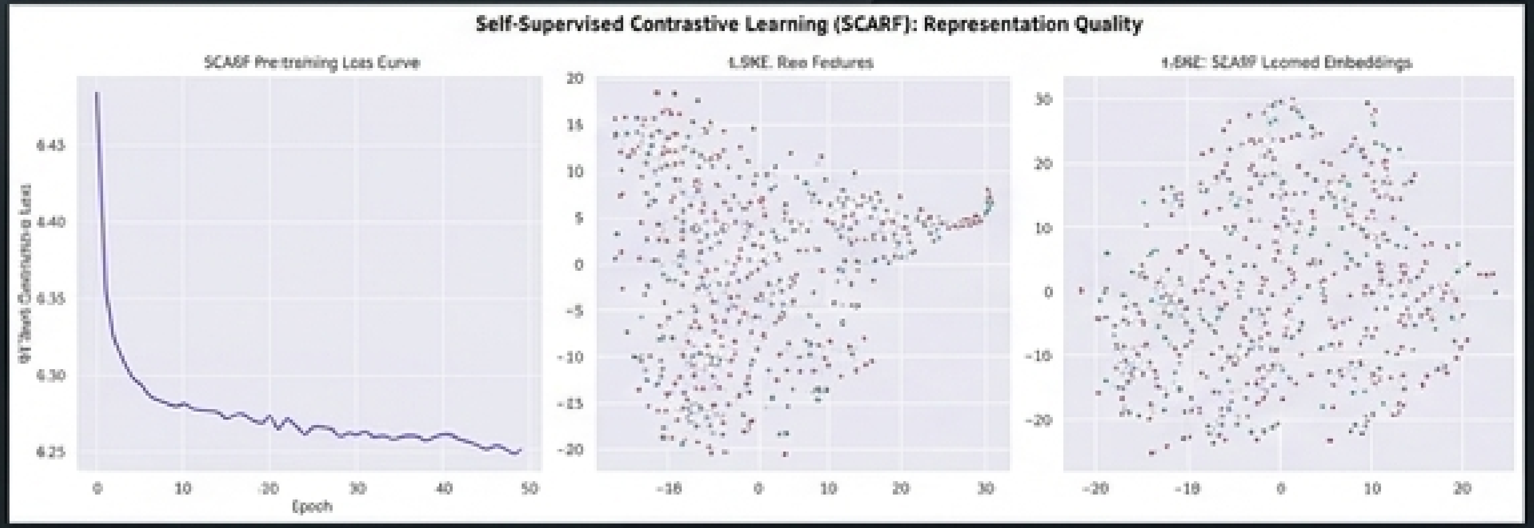
No mathematical combination of hyperparameters can extract a signal that doesn't exist.

SIGNAL ABSENCE CONFIRMED

Self-Supervised SCARF & Zero-Shot Transformers

Bypassing standard gradient descent to hunt for hidden latent structures.

A



B

Self-Supervised SCARF (Contrastive Pre-Training)

Forced to learn representations via random feature corruption and NT-Xent loss (temperature = 0.5).

Latent Reality: The t-SNE projection reveals successful and failed matches overlap entirely. The latent space is fundamentally inseparable.

TabPFN (Zero-Shot Tabular Transformer)

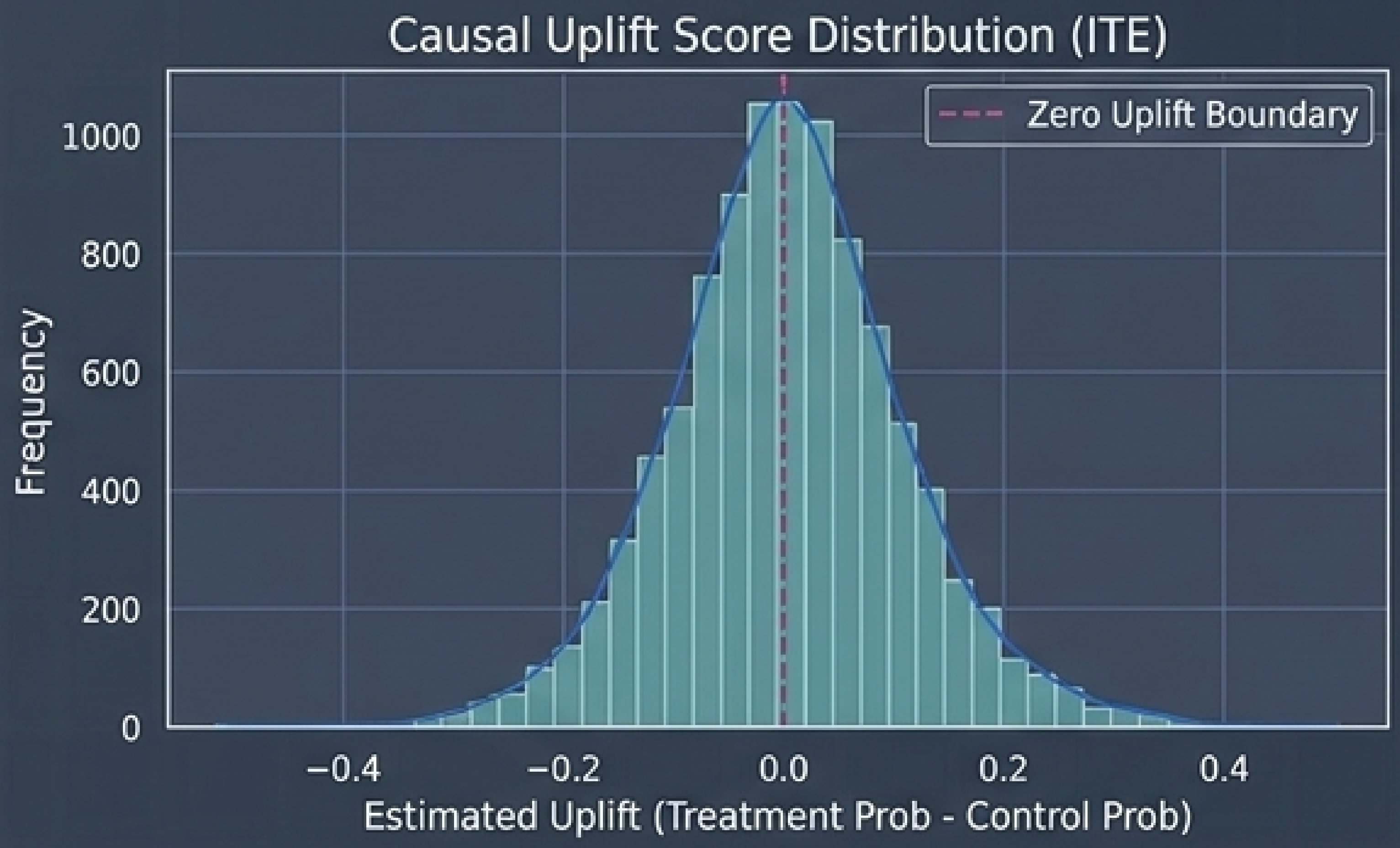
Pre-trained on millions of datasets to approximate the true Bayesian posterior.

Methodological Rigor: Evaluated strictly on a 1,000-sample computational subset; hybrid fallback dilution completely removed.

Result: Confirms SCARF—the dataset lacks separable geometric structure.

Causal Uplift Segment Targeting

Individual Treatment Effect (ITE)



Causal Prescriptive Marketing Matrix

DML Average Treatment Effect (ATE) = 0.0104 (p = 0.0322)

Conclusion: Causal uplift is statistically significant but structurally minimal. Intervention ceiling reached.

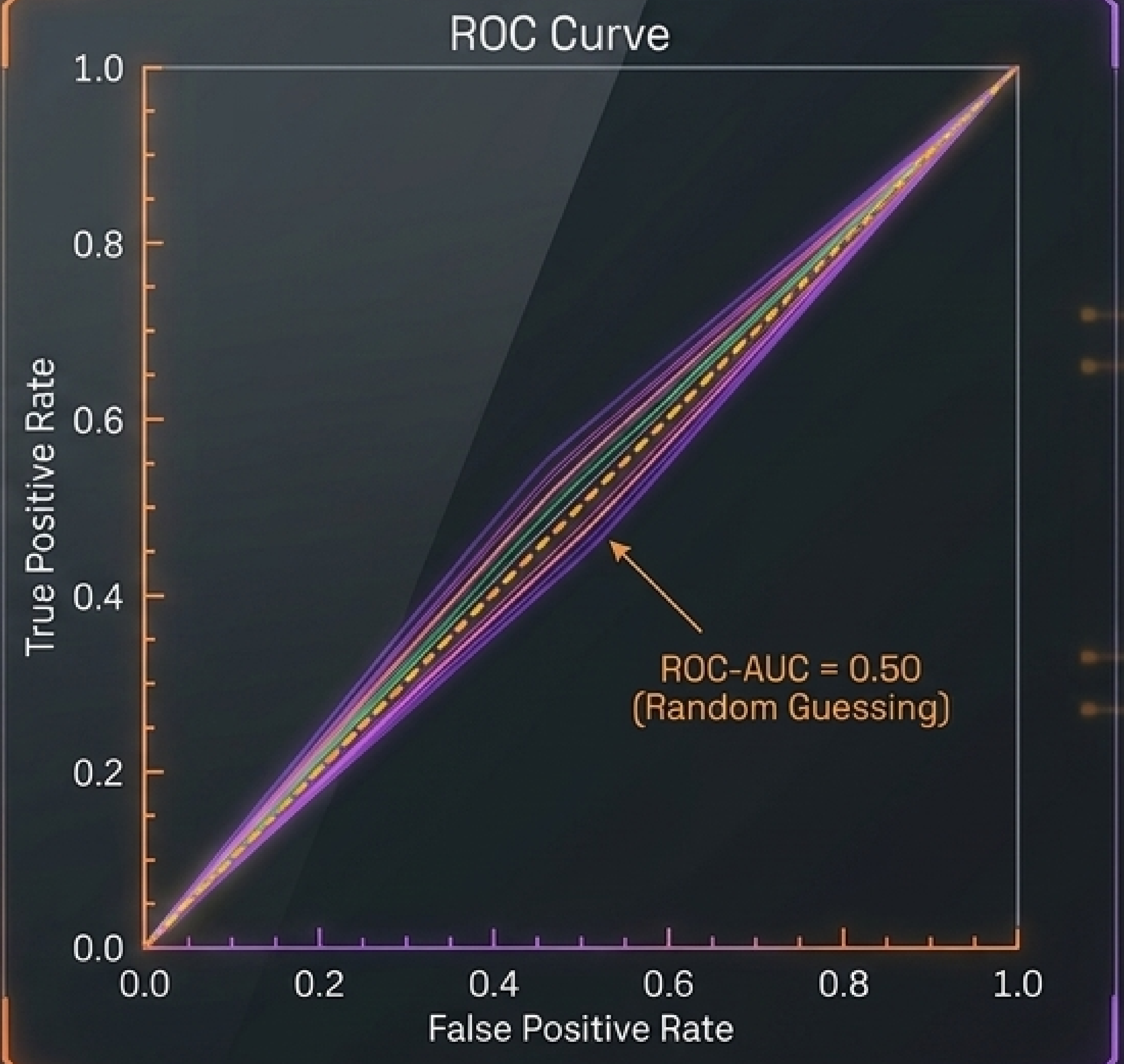
The Scientific Truth: Convergence at the Null Hypothesis

Perfect engineering reveals the empirical absence of predictive signal.

The Empirical Reality

All 16 models—despite SOTA architectures and 1,000-trial tuning—converge precisely at ~60.30% accuracy.

ROC-AUC is mathematically locked at ≈ 0.50 (Random Guessing).



The Scientific Twist: No Free Lunch

This is not a failure of Machine Learning. It is a mathematical proof of the dataset's nature.

Superficial behavioral features (swipe ratios, zodiac signs, usage time) carry zero genuine correlation with human romantic connection.

Our Stance: Scientific honesty demands we accept the data, rather than engineering an overfitted lie.

SwipeIQ V2 Dashboard & Strategic Recommendations

Engineering the Artifact

The Strategic Pivot

Sunset Tabular Metrics

Mathematical proof confirms demographic and tabular features lack genuine causal signal. Predictive ceiling capped at ~60%.

Pivot to NLP Bio Analysis

Capture the true, non-linear signals of human romantic connection by analyzing semantic depth and sentiment in user text.

Integrate Behavioral Latency

Transition matching algorithms to rely on active, in-app behavioral cues and temporal response sequences.